



# General Physics II

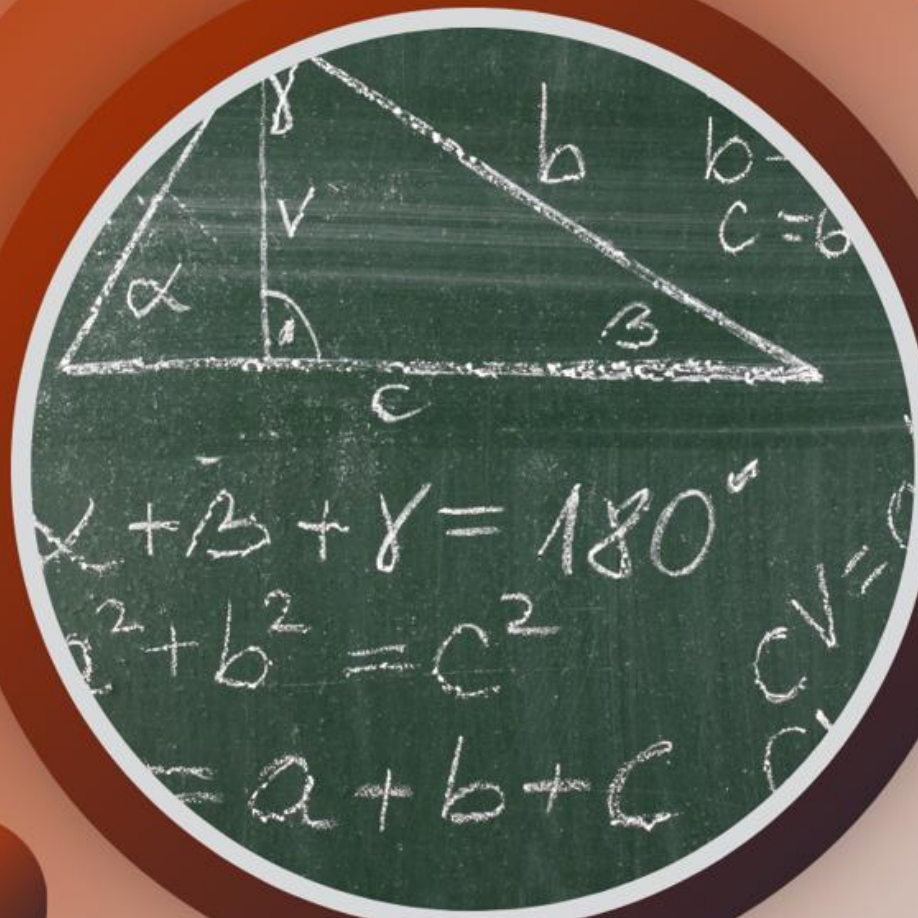
PHY121

*College of Science and Technology*



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# **MODULE 2: ELECTRIC CIRCUITS & MATERIALS**

## **UNIT 1: Electric Dipoles and Conductors/Insulators**

# Conductors and insulators

## Introduction

**Electrostatics deals with charges at rest** and the **electric fields** they produce.

In many physical systems, understanding how materials respond to electric fields is crucial for explaining phenomena such as:

- **Energy storage in capacitors**
- **Behavior of insulating materials**
- **Operation of electronic devices**

This lecture develops a logical progression from:

**Material response (conductors/insulators) → Microscopic behavior (dipoles) → Devices (capacitors) → Energy storage (electric fields).**

## Conductors and insulators Cont'd

### Properties of Conductors

#### 1. Free Charge Movement

- ☐ Electrons respond immediately to applied electric fields.
- ☐ Charges **redistribute until equilibrium** is reached.

#### 2. Electrostatic Equilibrium

At equilibrium:

- ☐ No net motion of charge
- ☐ Internal **electric field** becomes zero

#### 3. Charge Distribution

- ☐ Excess charge resides on the surface
- ☐ Higher concentration at sharp points

## Conductors and insulators Cont'd

### 4. Field Direction

- ❑ Electric field is perpendicular to the surface

### NOTES

The conduction of charges in conductors is governed by **Ohm's law**- states that the **current flowing** through a conductor is directly proportional to the potential difference across it, provided that temperature and other physical conditions remain constant.

Mathematically,

$$V \propto I \quad (1)$$

Introducing the constant of proportionality, we have:

$$V = IR \quad (2)$$

## Conductors and insulators Cont'd

Where  $V$  = Potential difference ( voltage applied in volts V);

$I$  = Current in Ampere A and

$R$  = Resistance in Ohms  $\Omega$

From equation (2), it follows that

- ❑ Increasing voltage increases current provided that  $R$  is constant.
- ❑ Resistance ( $R$ ) measures the opposition to the flow of current.
- ❑ Equation (2) or Ohm's law, applies to ohmic materials such as metals that **conduct** heat and **electricity**.
- ❑ The law is not applicable to semiconductors with negative temperature coefficient( i.e as temperature increases, resistance decreases, and so **conductivity increases**).

## Conductors and insulators Cont'd

Given

$$V = 2 \text{ V}$$

$$I = 2 \text{ mA} = 2 \times 10^{-3} \text{ A}$$

From Ohm's law  $V = IR$ ,

or 
$$R = \frac{V}{I} = \frac{2}{2 \times 10^{-3}}$$

$$R = 1000 \Omega$$

At 350K, given parameters:

$$V = 2 \text{ V}$$

$$I = 5 \text{ mA} = 5 \times 10^{-3} \text{ A}$$

Applying Ohm's law,

## Conductors and insulators Cont'd

$$R = \frac{V}{I} = \frac{2}{5 \times 10^{-3}}$$

$$R = 400 \, \Omega$$

It can be seen that resistance decreases as temperature increases, hence, the materials does not obey Ohm's law.

### Insulators

This is a material in which there is **no free charge** movement.

Common examples of insulators include:

- ☐ Glass
- ☐ Rubber
- ☐ Plastic

## Conductors and insulators Cont'd

### Polarization in insulator

Unlike in conductor where there is charge redistribution due to an **applied field**, when an electric field is applied to an insulator:

1. Charges shift slightly within atoms
2. Induced **dipole** is created

Definition, **polarization** is the process whereby an induced dipole is created in an insulator when to an applied electric field.

| Property          | Conductors     | Insulators   |
|-------------------|----------------|--------------|
| Charge mobility   | Free           | Bound        |
| Internal field    | Zero           | Non-zero     |
| Response to field | Redistribution | Polarization |

# Electric Dipole

## Electric dipole

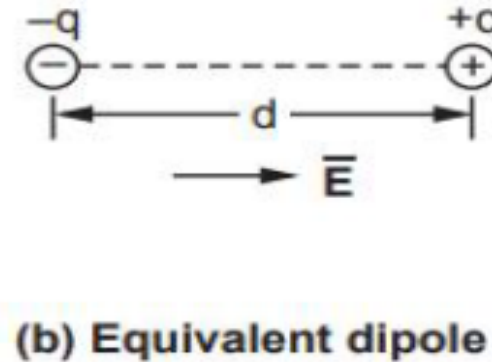
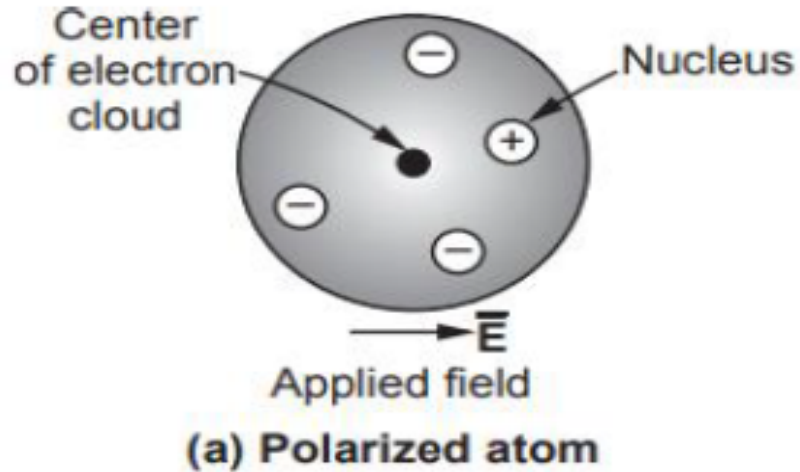


Figure 1. A schematic of polarization and electric dipole

Consider the system of two opposite charges shown in Fig 1b. It can be seen that an electron cloud has a centre separated from the nucleus. This forms an electric dipole. A system of two equal and opposite charges(+q and -q) separated by a *small* distance, d constitute an electric dipole.



## Electric Dipole Cont'd

### Dipole moment

This is defined as:

$$p = qd \quad (3)$$

Where

*$q$  = magnitude of the charge in Coulomb (C)*

*$d$  = separation distance in meter (m)*

**Class drill:**

**Given the above units for  $q$  and  $d$ , what is the unit for dipole moment?**

## **Electric Dipole Cont'd**

**Dipole moment is measured in Coulomb – metre (C·m)**

### **Dipole in an electric field**

**When placed in a uniform electric field, a torque is created that causes rotation and tends to align dipole with the applied field.**

**Mathematically,**

**Torque on dipole:**

$$\tau = pE \sin \theta \quad (4)$$

**Potential energy**

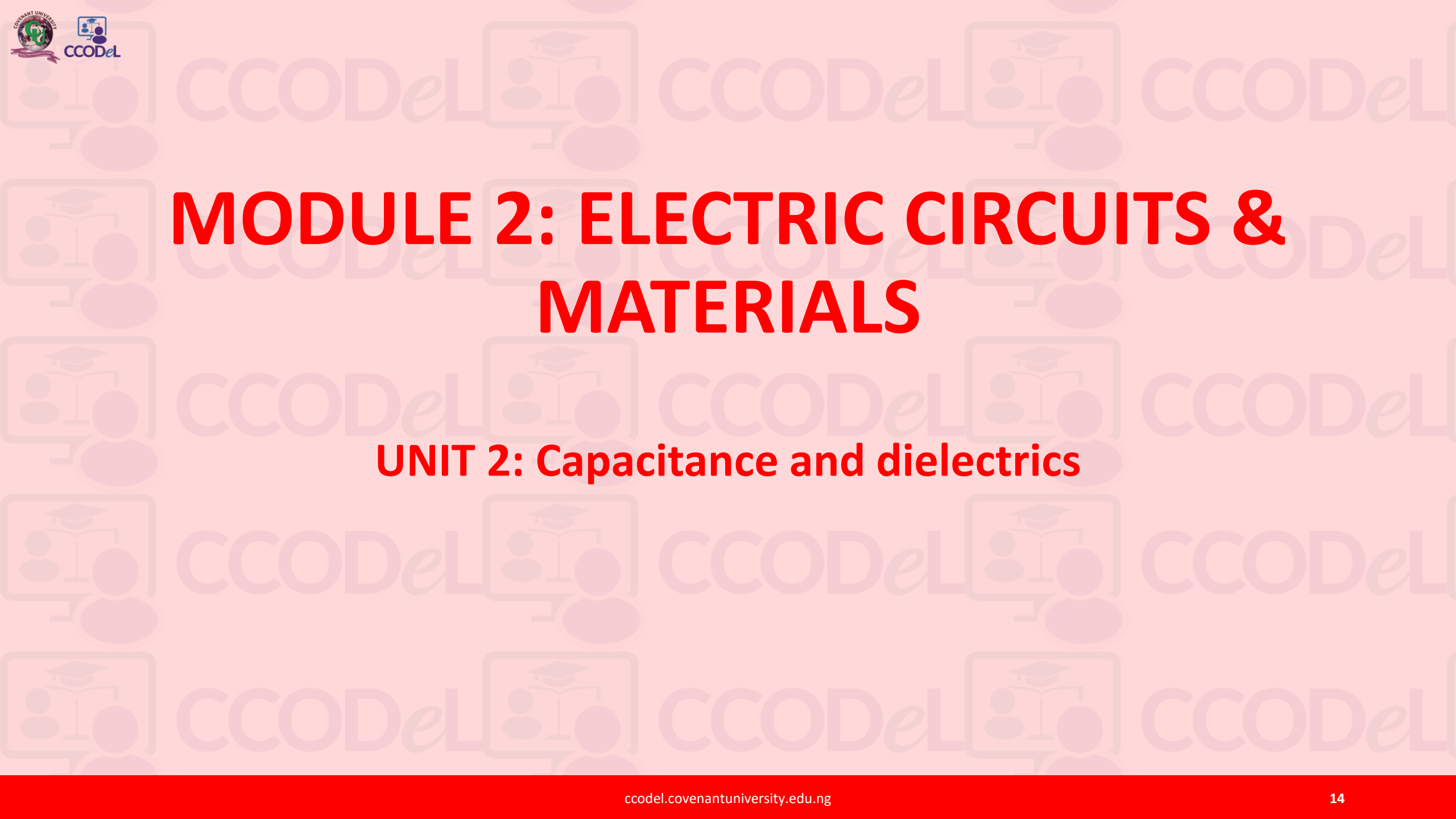
$$U = pE \cos \theta \quad (5)$$

## Electric Dipole Cont'd

Equation (5) shows there is an alignment of dipole with the field at minimum energy. At maximum energy, however, dipole is opposite the field.

### Electric Dipole and Insulators

*Polarization in insulators arises from many dipoles aligning*



# **MODULE 2: ELECTRIC CIRCUITS & MATERIALS**

## **UNIT 2: Capacitance and dielectrics**

# Capacitance

## Concept of Capacitance

A **capacitor** is an electronic device for storing **electrical energy** by allowing charges to accumulate on metal plates.

When a voltage is applied across a capacitor:

- ❑ One plate becomes positively charged.
- ❑ The other plate becomes negatively charged.

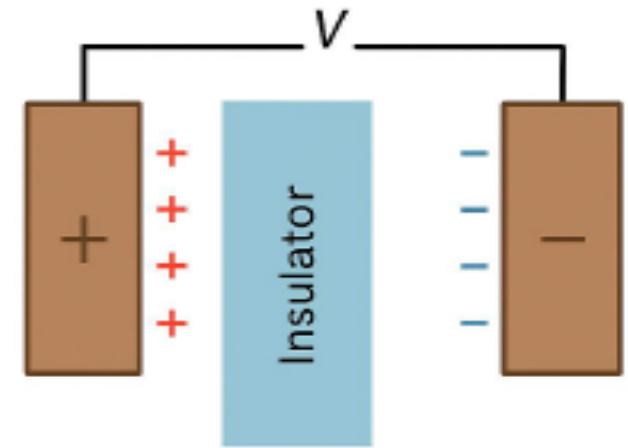


Figure 2. Parallel-Plate Capacitor

## Capacitance Cont'd

- ❑ The **electric field** between the plates stores **electrostatic energy**.

This **electrical energy** is recovered when these **charges** are allowed to move away from these plates into the circuit of which the capacitor forms a part.

Generally, any device which can store charge is a capacitor.

### Definition

The capacitance  $C$  of a **capacitor** is defined as the ratio of the magnitude of charge stored  $Q$  on either conductor to the potential difference  $V$  between them:

## Capacitance Cont'd

$$C = \frac{Q}{V} \quad (6)$$

The SI unit of capacitance is farad (F). Other units are simply multiple of F, i.e micro farad ( $\mu F = 10^{-6} F$ ), pico farad ( $pF = 10^{-12} F$ ), etc.

Note that in determining the capacitance of a capacitor, the geometry of the surface storing the charge is considered. This is because potential is **geometrically** dependent.

### Example 1

Consider an insulated conducting sphere of shell radius  $R$  that can store charge. If a charge  $Q$  is placed on it, the outer surface of the shell becomes an equipotential surface. Determine:

1. Potential of the outer surface and
2. Capacitance of the shell.

**Solution:**

1. For a conducting sphere, all charge resides on the surface and behaves like a point charge at the center (for external points). Thus, the potential at the surface is:

$$V = \frac{Q}{4\pi\epsilon_0 R}$$

This is constant over the surface, hence it is an equipotential surface.

Note that potential is zero as  $R \rightarrow \infty$ . Here, we can take the ground as zero potential.

## 2. The capacitance

$$C = \frac{Q}{V}$$

But  $V = \frac{Q}{4\pi\epsilon_0 R}$  and so,

$$C = \frac{Q}{V} = 4\pi\epsilon_0 R \text{ Coulomb/volt OR}$$

$4\pi\epsilon_0 R$  F since 1 farad = Coulomb/volt.

- ❑ As can be inferred from the above, **capacitance ( $C$ ) depends on geometry( $R$ ).**
- ❑ Larger sphere gives larger capacitance.

## Capacitance Cont'd

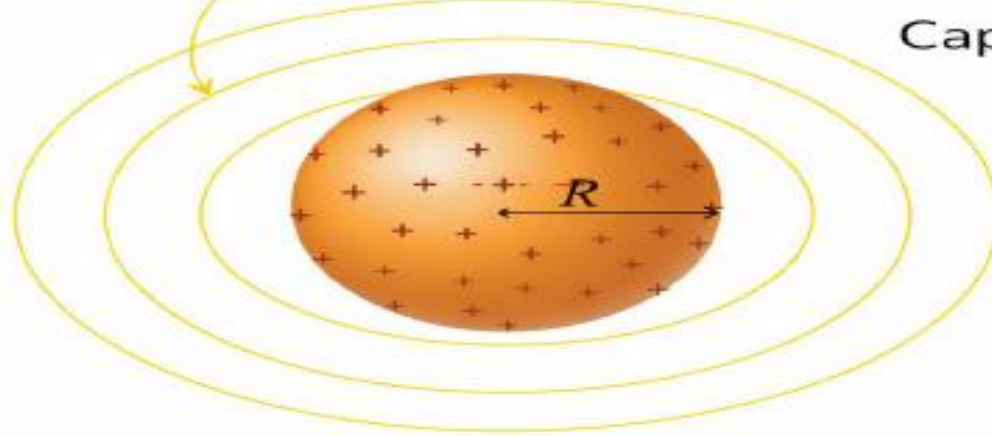
### Conducting Spherical Shell: Potential and Capacitance

$$\infty V = 0$$

$$V = \frac{Q}{4\pi\epsilon_0 R}$$

$$\text{Potential: } V = \frac{Q}{4\pi\epsilon_0 R}$$

$$\text{Capacitance: } C = 4\pi\epsilon_0 R$$



### Class work

**Determine the value of the capacitance if the radius of the spherical shell is  $R = 100 \text{ cm}$ . Take  $\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$ .**

**Answer to Class work**

$$C = 1.1 \times 10^{-10} F$$

### Parallel Plate Capacitor

**This is the most commonly used capacitor. The symbolic representation of a capacitor is**



**It consists of two metal sheets or plates (circular, cylindrical or rectangular) kept close to each other with an insulator separating the two sheets. See Fig 2.**

**This system has the ability to have larger capacity without having the corresponding larger dimensions.**

## Capacitance Cont'd

The capacitance of a parallel plate capacitor is given as:

$$C = \frac{\epsilon_0 A}{d} \quad (7)$$

From equation (7), the factors that affect the capacitance of a parallel plate capacitor include:

1. Area of plate.
2. Separation between plates.
3. Dielectric materials insulating the plates.

Dielectric increases the capacitance of a parallel plate capacitor by a factor called dielectric constant  $\epsilon_r$ .

## Capacitance Cont'd

**Relation between capacitance and dielectric constant:**

$$C = \epsilon_r C_o \quad (8)$$

Where  $C_o$  = *capacitance in vacuum, ie no dielectric.*

### **Example 2**

**A parallel plate capacitor has capacitance  $8.85 \times 10^{-11} F$ . What is the capacitance if a dielectric material with dielectric constant of 3 is inserted between the plates?**

### **Solution**

**Here, the capacitance before dielectric is**

$$C_o = 8.85 \times 10^{-11} F$$

## Capacitance Cont'd

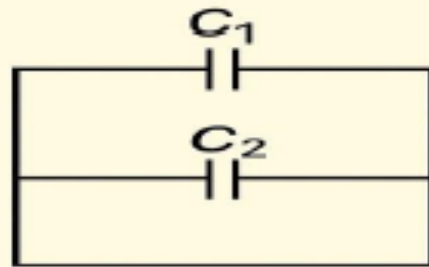
Hence, applying dielectric,

$$C = \epsilon_r C_o = 3 \times 8.85 \times 10^{-11} F$$

$$C = 2.7 \times 10^{-10} F$$

### COMBINATIONS OF CAPACITORS

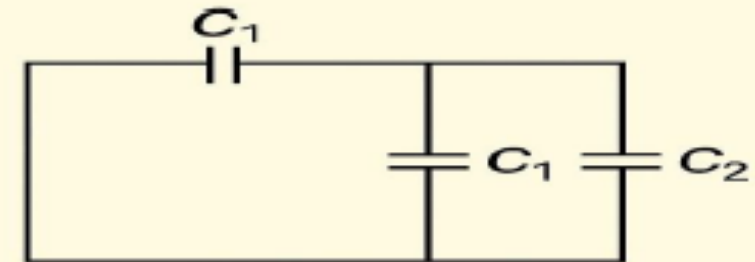
#### PARALLEL



$$C_{eq} = C_1 + C_2$$

$$V_1 = V_2 = V$$

#### SERIES



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}, \text{ where } C_p = C_1 + C_2$$

$$Q_1 = Q_2 = Q$$

## Capacitance Cont'd

### Characteristics of Parallel Combination

**1. Capacitances add:**

$$C_{eq} = C_1 + C_2 + \dots + C_n \quad (9)$$

**2. Voltage across each is the same:  $V_1 = V_2 = V$ .**

**3. Total charge is sum of charges:  $Q_{tot} = \sum C_i V$**

### Characteristics of Series (end-to-end connection) Combination

**1. Equivalent capacitance is inverse addition:**

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n} \quad (10)$$

**2. Charge on every capacitor is the same:  $Q_1 = Q_2 = Q$ .**

## Capacitance Cont'd

**3. Voltage divides:**  $V_i = \frac{Q}{C_i}$  and  $V = \sum_i V_i$

**4.  $C_{eq} < \min(C_i)$ , that is, the equivalent capacitance is less than the smallest capacitor in the combination, i.e. series reduces capacitance.**

**Energy Stored in Capacitor**

**Energy Stored in a Capacitor**

**Consider charging a capacitor from *zero charge up to charge  $Q$* .  
At an intermediate stage the capacitor carries charge  $q$  and the potential difference across it is:**

$$v = \frac{q}{C} \quad (11)$$

## Capacitance Cont'd

To add an infinitesimal extra charge  $dq$ , there must be an infinitesimal amount of work  $dW$  done against the electric potential:

$$dW = v dq \quad (12)$$

(Recall that we define *potential difference*  $V$  as the work done  $W$  in moving a charge  $Q$  from one point A to another point B)

$$dW = \frac{q}{C} dq.$$

- Total work (energy) required to charge the capacitor from 0 to  $Q$  is

$$W = \int_0^Q \frac{q}{C} dq = \frac{1}{C} \int_0^Q q dq \quad (13)$$

## Capacitance Cont'd

$$W = \int_0^Q \frac{q}{C} dq = \frac{1}{C} \int_0^Q q dq \quad (13)$$

$$W = \frac{1}{C} \left[ \frac{q^2}{2} \right]_0^Q = \frac{1}{C} \left( \frac{Q^2}{2} - 0 \right)$$

$$= \frac{1}{C} \frac{Q^2}{2} = \frac{Q^2}{2C} \quad (1.23)$$

**Using  $Q = CV$  where  $V$  is the final voltage, other forms of the energy stored in a capacitor can be obtained thus:**

$$W = \frac{Q^2}{2C} = \frac{(CV)^2}{2C} = \frac{V^2 C}{2} = ??$$

## Capacitance Cont'd

Answer

$$W = \frac{V^2 Q}{2V} = \frac{QV}{2}$$

## Capacitance Cont'd

### Example 3

A circuit has two capacitors  $C_1 = 2.0\mu F$  and  $C_2 = 3.0\mu F$  in parallel. This parallel combination is in series with  $C_3 = 4.0\mu F$ . If the entire combination is connected to a 12 V battery, find:

- The equivalent capacitance of the combination
- Total charge across the circuit
- Total energy stored

## Capacitance Cont'd

**Solution:**

**a. Parallel combination:**

$$C_p = C_1 + C_2 = 2 + 3 = 5\mu F$$

**$C_p$  combines with  $C_3$  in series:**

**The equivalent capacitance of the combination is**

$$\frac{1}{C_{eq}} = \frac{1}{C_p} + \frac{1}{C_3} \text{ or}$$

$$C_{eq} = \frac{5 \times 4}{5 + 4} = \frac{20}{9} = 2.225\mu F$$

**b. Total charge:**

$$Q = C_{eq} \cdot V = 2.22 \times 10^{-6} \times 12 = 26.6\mu C$$

**c. Total energy stored:**

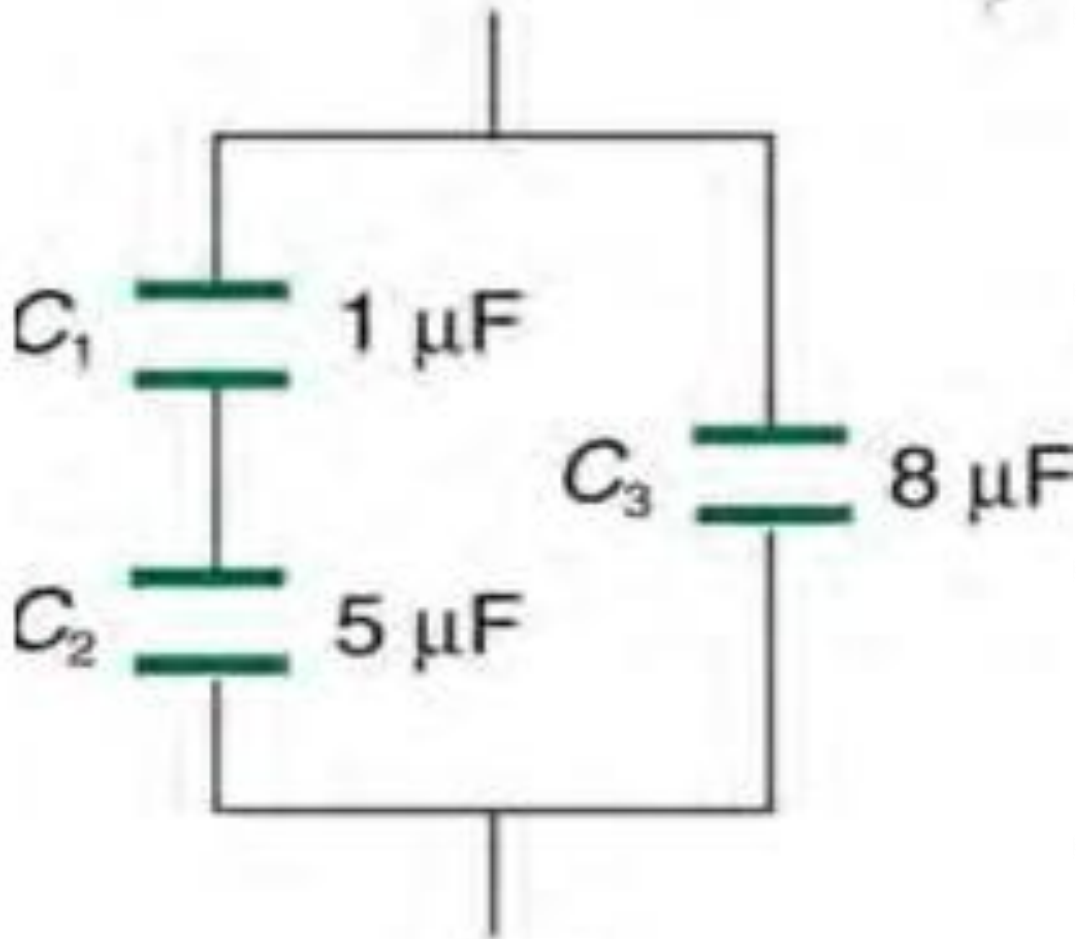
## Capacitance Cont'd

**Total energy stored**

$$W = \frac{CV^2}{2} = 1.60 \times 10^{-4} J$$

# Classwork

1. Determine the total capacitance of the capacitors



2. What is the capacitance of a capacitor having a stored energy of 2.25 mJ after being charged with a 30-volt source?
3. The area of each plate of a parallel plate capacitor is  $4 \times 10^{-2} \text{ m}^2$ . If the thickness of the dielectric placed in between the plates is  $10^{-3} \text{ m}$  and the relative permittivity of the dielectric is 7, then the capacitance of the capacitor is?
4. Given a circuit where  $R_1$ ,  $R_2$  and  $R_3$  are in series with a voltage source of 48 V, such that  $R_1 = 7 \Omega$ ,  $R_2 = 6 \Omega$  and  $R_3 = 3 \Omega$ , calculate the current that flows through  $R_3$ ?
5. A metal sphere mounted on an insulating rod carries a charge of 4.5 nC when its potential is 180 V higher than its surroundings. What is the capacitance of the capacitor formed by the sphere and its surroundings?